

PORTFOLIO

AUGUST 2026

LIAM O'CONNELL

ABOUT ME

I am a junior Mechanical Engineering student at the University of Maryland. I have extensive experience in CAD, engineering analysis, hardware development, and am particularly interested in robotics and the opportunity to integrate AI into engineering workflows.

Contact:

267-449-2202

loconne2@terpmail.umd.edu

www.linkedin.com/in/liamoconnellumd



Professional Experience

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Calculations Demo

DEFLECTION

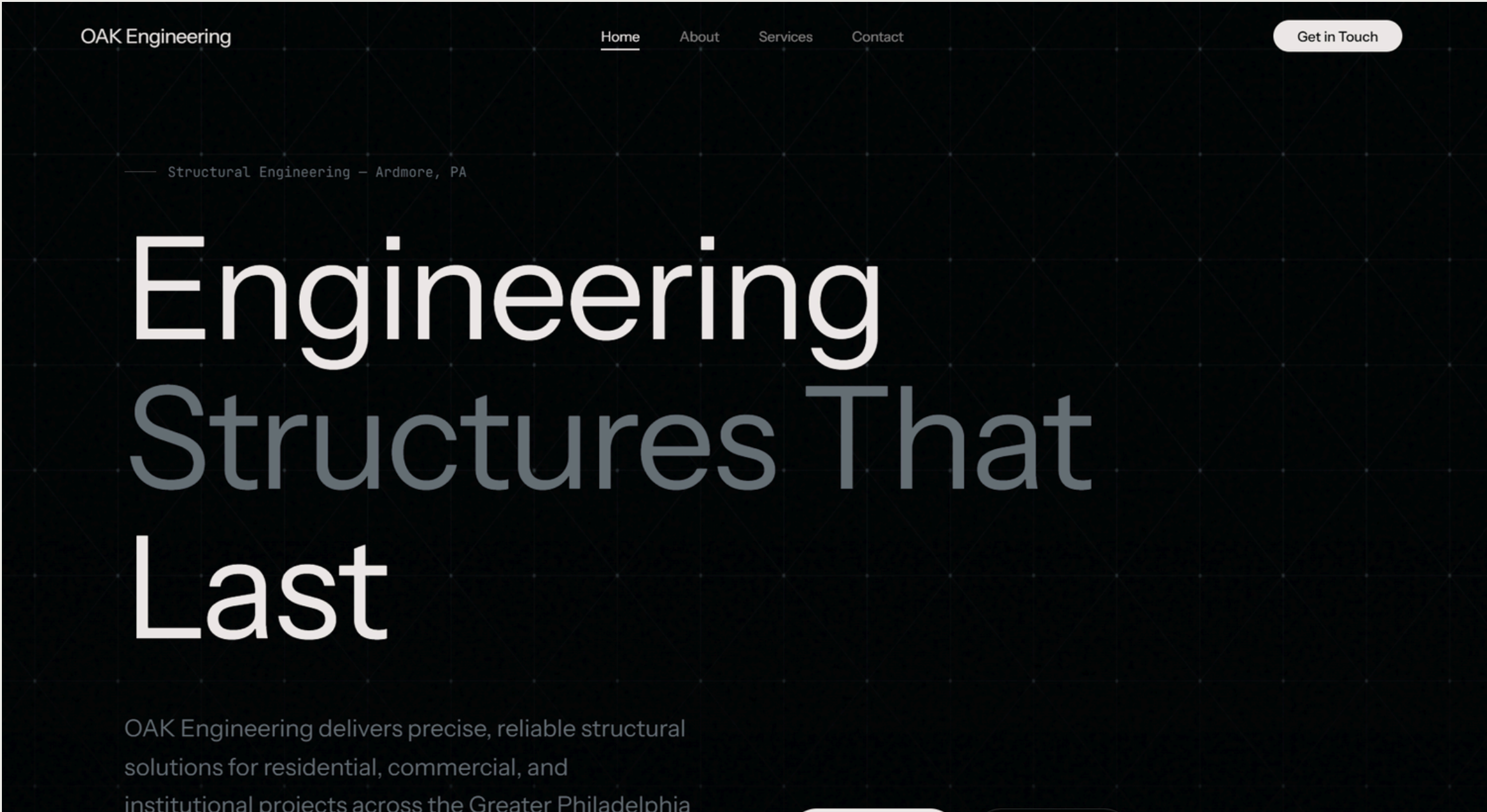
Point Load (lbs):	300
Length (inches):	16
Radius (inches):	0.5
Young's Mod (psi):	29,000,000
Deflection (inches):	0.0179839
Check:	OK

BENDING MOMENT (SIDE RAIL)

Point Load (lbs):	300
Length (inches):	60
Base (inches):	3/8
Height (inches):	3
Moment (lb*in):	4,500
Inertia (in^4):	5/6
Max Stress (ksi):	8
Check:	OK

AXIAL BUCKLING (SIDE RAIL)

	Lateral	In/OOP
Point Load (lbs):	600	600
Length (Lc, inches):	16	60
Height (inches):	3/8	3
Base (inches):	3	3/8
Young's Mod (psi):	29,000,000	29,000,000
Inertia (in^4):	0	5/6
Area (in^2):	1 1/8	1 1/8
Radius of Gyr (in):	0.1082532	0.866025
Max Stress (ksi):	6.8409935	31.13371
Actual Stress (ksi):	1/2	1/2
Check:	OK	OK



This past summer at Oak Engineering (a local structural engineering firm), I performed first principles structural analysis on live projects, sizing and verifying steel members against bending, buckling, and deflection limits to evaluate designs and confirm they carry their loads safely. The sheet shown is a representative example of that work a fixed-ladder side-rail analysis run against a defined design load according to OSHA requirements. I also assisted with AI integration – I designed and built Oak's production website end to end, owning the architecture and deployment (via Cloudflare), including an interactive tool that structures and routes inbound client requests.

Terrapin Rocket Team

AUGUST 2026

I am a member of the Terrapin Rocket Team at the University of Maryland. I am on the aerostructures subteam, where my work spans across multiple different areas. I use CAD and FEA to analyze and design structural components. Furthermore, I develop testing parameters to validate structures and inform design decisions between different compositions. I participate in running these tests and interpreting their results.

I also use RocketPy and OpenRocket to help simulate conditions and analyze performance. This helps with verifying that the rocket will be able to reach its targeted launch altitude of 30,000 feet, as well as to independently analyze specific parts of the rocket under their respective stresses. Our team meets weekly for performance updates and task delegation.



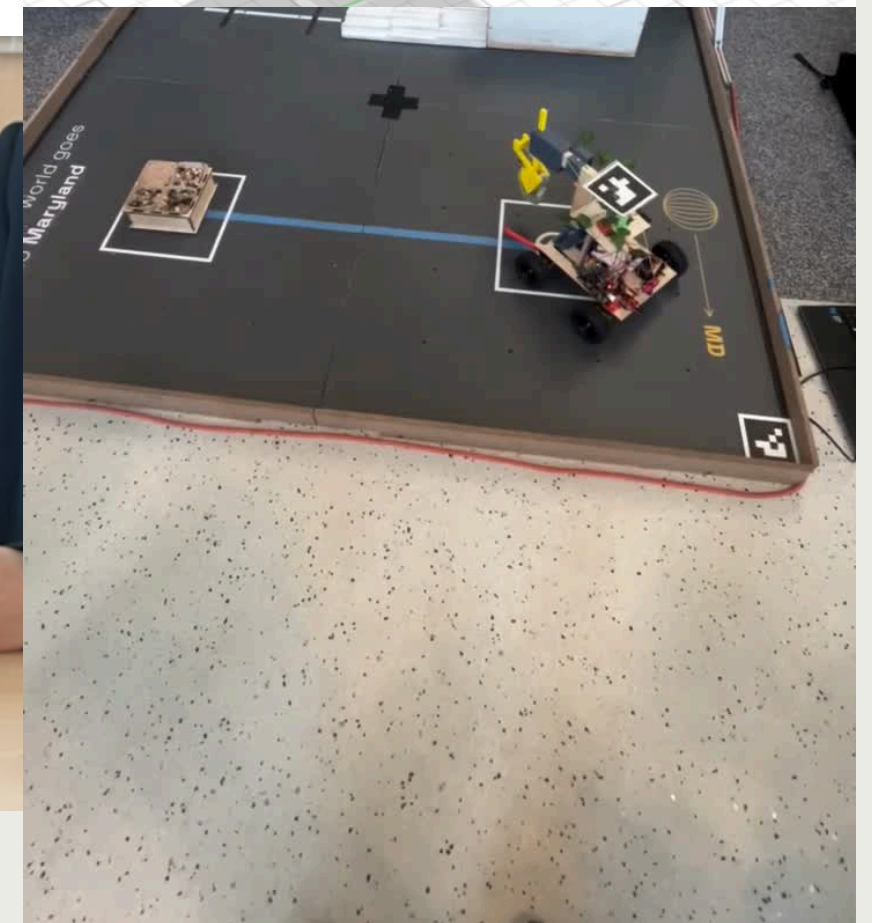
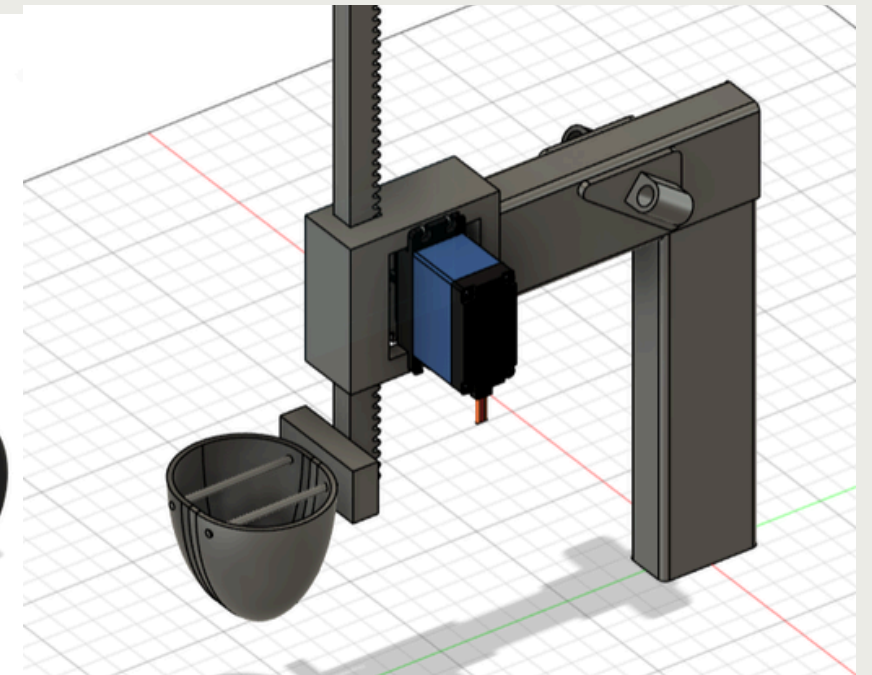
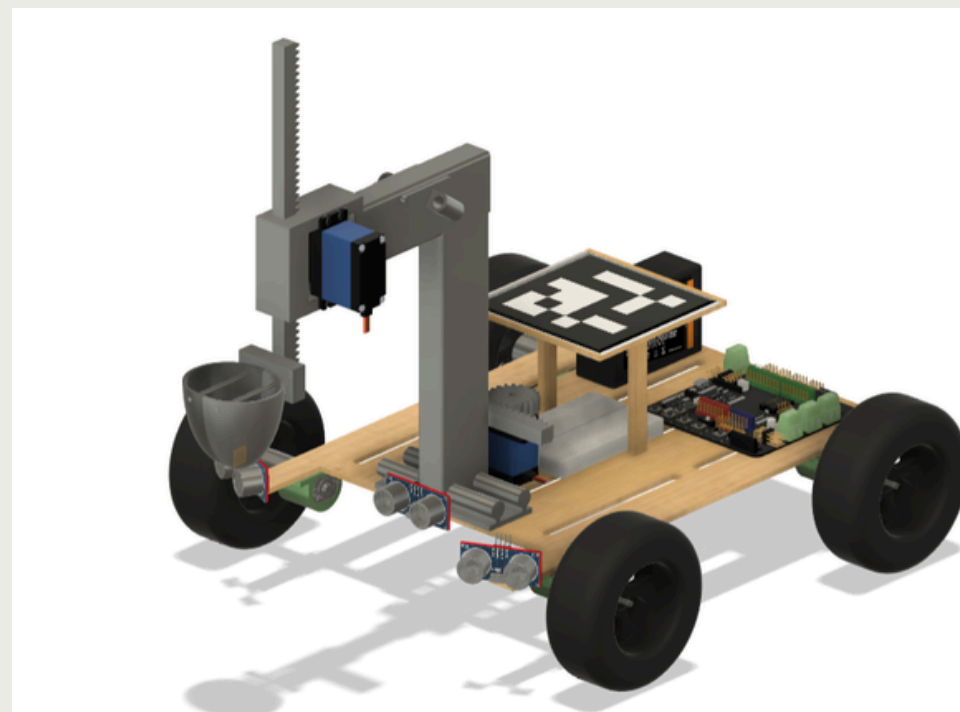
OTV Project

Designed and built a fully autonomous vehicle that navigated a randomized obstacle arena, reached a mission site, identified a target material, collected a sample, and returned without any operator input.

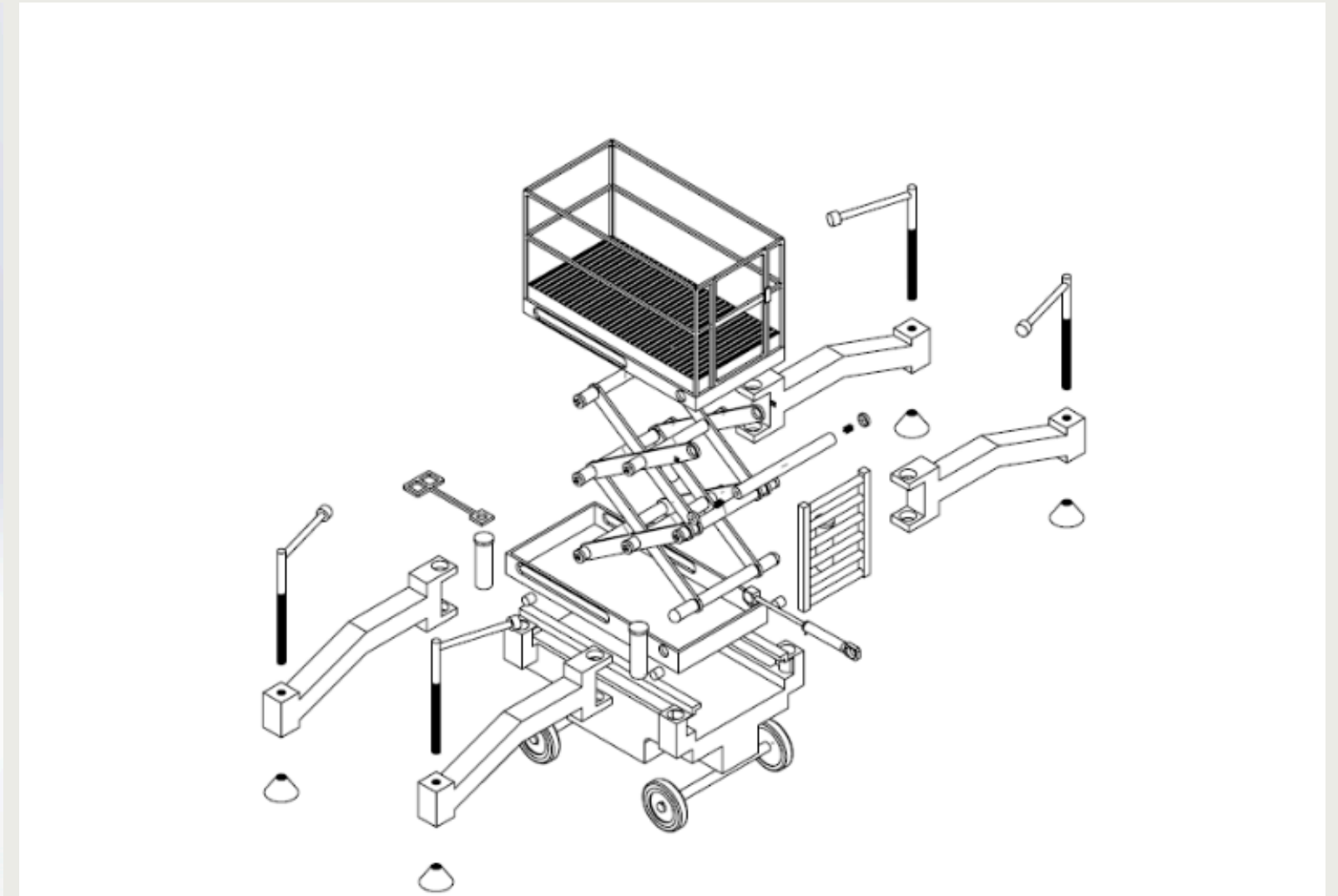
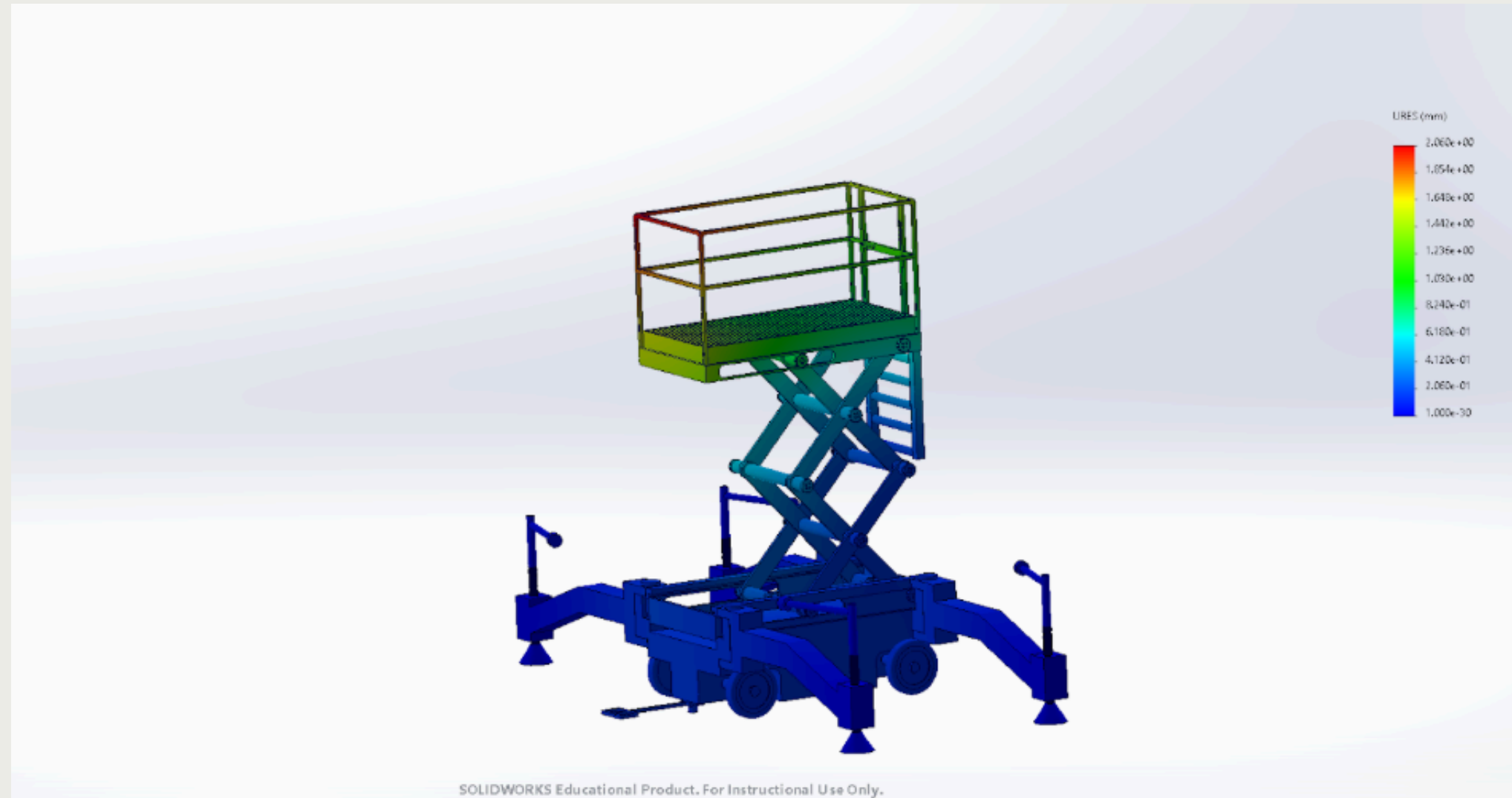
As mechanical sub-team lead, I owned the sample-collection mechanism end to end. I designed an arm driven by servos and linear actuators that inserted into the material bed, opened to scoop its contents, and closed to retain the sample, then used a load cell to identify the correct material by measured weight.

I modeled the mechanism and packaging via CAD, using assemblies to resolve fitment, mounting, and interface constraints across the drive system, sensors, and mission hardware, and refined the design through 3D-printed revisions and repeated testing across hundreds of iterations.

On the strength of our team's performance, we were invited to a private showcase to demonstrate the vehicle live for faculty and engineering leadership.



Scissor Lift - Design & Analysis



Designed and analyzed a complete scissor lift assembly in SolidWorks as CAD & FEA lead on a six-person team. I owned the full finite element analysis — interpreting stress, displacement, and yield margins across the linkage to establish a 500 kg rated load capacity at a factor of safety of 3 — and applied GD&T across all mating components. The exploded view and bill of materials document the full assembly, and a motion study validated the linkage's range of travel through a complete lift cycle.

Technical Skills

- I have extensive project experience using **Siemens NX** and **Autodesk Fusion**.
- I have a certification (CSWA) for my **SolidWorks** ability.
- I am experienced with and have coursework in **Python**, **Numpy**, and other methods of **data analysis**.
- **Adept with AI** – I developed an agentic AI based engineering proposal development tool for my internship last summer.
- **Relevant coursework** in Thermodynamics, Fluid Mechanics, Materials Engineering, Statics, and Physics.
- Experience with **engineering calculations** (produced multiple calculations packages for commercial projects over the past summer at my internship) and **engineering drawings**, which I also made on site visits.
- I have extensive **experience with hand and power tools**, honed through my project experience.

- My full list of technical proficiencies:

CAD & Analysis: Creo, SolidWorks (CSWA), FEA, ANSYS, HyperMesh, Siemens NX, Fusion 360, GD&T

Programming & Data Analysis: Python, MATLAB, NumPy, Matplotlib, JavaScript, Git/GitHub, Sensor Integration, C++

Fabrication & Prototyping: 3D Printing, Laser Cutting, Soldering, Circuit Wiring, Component Assembly, Rapid Prototyping

Thank You

Thank you for taking the time to review my portfolio. I appreciate the opportunity to share my project experience and technical background. I am eager to continue developing my skills and contributing in a hands-on engineering environment. If you have any questions, please feel free to contact me below:

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